OpenEdge Bridge — Documentation

Sample INSERT Data

MySQL — `id_print` (11 records)

```
INSERT INTO id_print (batch_no, batch_date, app_no, dob, cp_no, fullname, hcp_fullname, address, hcp_cpno, lname, fname, mname, spa_docno, page_no, book_no, series_no, spa_atty, spa_dt_issue, printed, printed_dt, printed_tm, tran_type, image_data, id_code, sfx, photo, gender, date_issued, id_no, year_applied, control_no, print_count) VALUES (1, '2026-01-15', 1001, '1990-05-20', 'CP-001', 'Juan Dela Cruz', 'Dr. Maria Santos', '123 Rizal St, Manila', 'HCP-001', 'Dela Cruz', 'Juan', 'Santos', 'SPA-001', 'P001', 'BK001', 'SR001', 'Atty. Reyes', '2025-12-01', 'no', NULL, NULL, 1, NULL, 5001, '', NULL, 1, '2026-01-15', 10001, 2026, 1, 0), (1, '2026-01-15', 1002, '1985-08-12', 'CP-002', 'Maria Clara Reyes', 'Dr. Jose Rizal', '456 Mabini St, Quezon City', 'HCP-002', 'Reyes', 'Maria', 'Clara', 'SPA-002', 'P002', 'BK001', 'SR002', 'Atty. Cruz', '2025-12-02', 'no', NULL, NULL, 1, NULL, 5002, '', NULL, 2, '2026-01-15', 10002, 2026, 2, 0), (1, '2026-01-16', 1003, '1992-03-25', 'CP-003', 'Pedro Penduko', 'Dr. Ana Lopez', '789 Bonifacio Ave, Cebu', 'HCP-003', 'Penduko', 'Pedro', 'Garcia', 'SPA-003', 'P003', 'BK002', 'SR003', 'Atty. Santos', '2025-12-03', 'no', NULL, NULL, 2, NULL, 5003, 'JR', NULL, 1, '2026-01-16', 10003, 2026, 3, 1), (2, '2026-02-10', 1004, '1988-11-30', 'CP-004', 'Ana Patricia Gomez', 'Dr. Luis Torres', '321 Aguinaldo Rd, Davao', 'HCP-004', 'Gomez', 'Ana', 'Patricia', 'SPA-004', 'P004', 'BK002', 'SR004', 'Atty. Reyes', '2025-12-10', 'yes', '2026-02-11', '10:30:00', 1, NULL, 5004, '', NULL, 2, '2026-02-10', 10004, 2026, 4, 1), (2, '2026-02-10', 1005, '1995-07-14', 'CP-005', 'Carlos Miguel Tan', 'Dr. Elena Villanueva', '654 Luna St, Makati', 'HCP-005', 'Tan', 'Carlos', 'Miguel', 'SPA-005', 'P005', 'BK003', 'SR005', 'Atty. Dela Rosa', '2025-12-15', 'no', NULL, NULL, 3, NULL, 5005, '', NULL, 1, '2026-02-10', 10005, 2026, 5, 2), (2, '2026-02-11', 1006, '1982-01-08', 'CP-006', 'Sofia Isabelle Lim', 'Dr. Ramon Bautista', '987 Silang Ave, Pasig', 'HCP-006', 'Lim', 'Sofia', 'Isabelle', 'SPA-006', 'P006', 'BK003', 'SR006', 'Atty. Mercado', '2025-12-20', 'no', NULL, NULL, 1, NULL, 5006, '', NULL, 2, '2026-02-11', 10006, 2026, 6, 0), (3, '2026-03-05', 1007, '1998-09-18', 'CP-007', 'Rafael Antonio Uy', 'Dr. Patricia Ong', '159 Roxas Blvd, Paranaque', 'HCP-007', 'Uy', 'Rafael', 'Antonio', 'SPA-007', 'P007', 'BK004', 'SR007', 'Atty. Garcia', '2026-01-05', 'yes', '2026-03-06', '14:15:00', 2, NULL, 5007, '', NULL, 1, '2026-03-05', 10007, 2026, 7, 1), (3, '2026-03-05', 1008, '1991-12-03', 'CP-008', 'Gabriela Nicole Sy', 'Dr. Fernando Aquino', '753 Quezon Blvd, Taguig', 'HCP-008', 'Sy', 'Gabriela', 'Nicole', 'SPA-008', 'P008', 'BK004', 'SR008', 'Atty. Ramos', '2026-01-10', 'no', NULL, NULL, 1, NULL, 5008, '', NULL, 2, '2026-03-05', 10008, 2026, 8, 0), (3, '2026-03-06', 1009, '1987-04-22', 'CP-009', 'Miguel Alfonso Co', 'Dr. Teresa Javier', '246 Osmena St, Baguio', 'HCP-009', 'Co', 'Miguel', 'Alfonso', 'SPA-009', 'P009', 'BK005', 'SR009', 'Atty. Villar', '2026-01-15', 'no', NULL, NULL, 3, NULL, 5009, 'III', NULL, 1, '2026-03-06', 10009, 2026, 9, 3), (4, '2026-04-20', 1010, '1993-06-11', 'CP-010', 'Isabella Marie Chua', 'Dr. Roberto Domingo', '135 Kalayaan Ave, Pampanga', 'HCP-010', 'Chua', 'Isabella', 'Marie', 'SPA-010', 'P010', 'BK005', 'SR010', 'Atty. Pineda', '2026-02-01', 'no', NULL, NULL, 1, NULL, 5010, '', NULL, 2, '2026-04-20', 10010, 2026, 10, 0), (4, '2026-04-20', 1011, '1980-10-05', 'CP-011', 'Antonio Luis Pangilinan', 'Dr. Carmen Salazar', '864 EDSA, Mandaluyong', 'HCP-011', 'Pangilinan', 'Antonio', 'Luis', 'SPA-011', 'P011', 'BK006', 'SR011', 'Atty. Hernandez', '2026-02-10', 'yes', '2026-04-21', '09:00:00', 2, NULL, 5011, '', NULL, 1, '2026-04-20', 10011, 2026, 11, 2);
```

OpenEdge (Progress) — `pub."id-print"` (11 records)

```
INSERT INTO pub."id-print" ("batch-no", "batch-date", "app-no", "dob", "cp-no", "fullname", "hcp-fullname", "address", "hcp-cpno", "lname", "fname", "mname", "spa-docno", "page-no",
```

```
"book-no", "series-no", "spa-atty", "spa-dt-issue", "printed", "printed-dt", "printed-tm",
"tran-type", "image-data", "id-code", "sfx", "photo", "gender", "date-issued", "id-no",
"year-applied", "control-no") VALUES (1, '2026-01-15', 1001, '1990-05-20', 'CP-001', 'Juan Dela
Cruz', 'Dr. Maria Santos', '123 Rizal St, Manila', 'HCP-001', 'Dela Cruz', 'Juan', 'Santos',
'SPA-001', 'P001', 'BK001', 'SR001', 'Atty. Reyes', '2025-12-01', 0, NULL, NULL, 1, NULL, 5001,
'', NULL, 1, '2026-01-15', '10001', 2026, 1), (1, '2026-01-15', 1002, '1985-08-12', 'CP-002',
'Maria Clara Reyes', 'Dr. Jose Rizal', '456 Mabini St, Quezon City', 'HCP-002', 'Reyes', 'Maria',
'Clara', 'SPA-002', 'P002', 'BK001', 'SR002', 'Atty. Cruz', '2025-12-02', 0, NULL, NULL, 1, NULL,
5002, '', NULL, 2, '2026-01-15', '10002', 2026, 2), (1, '2026-01-16', 1003, '1992-03-25',
'CP-003', 'Pedro Penduko', 'Dr. Ana Lopez', '789 Bonifacio Ave, Cebu', 'HCP-003', 'Penduko',
'Pedro', 'Garcia', 'SPA-003', 'P003', 'BK002', 'SR003', 'Atty. Santos', '2025-12-03', 0, NULL,
NULL, 2, NULL, 5003, 'JR', NULL, 1, '2026-01-16', '10003', 2026, 3), (2, '2026-02-10', 1004,
'1988-11-30', 'CP-004', 'Ana Patricia Gomez', 'Dr. Luis Torres', '321 Aguinaldo Rd, Davao',
'HCP-004', 'Gomez', 'Ana', 'Patricia', 'SPA-004', 'P004', 'BK002', 'SR004', 'Atty. Reyes',
'2025-12-10', 1, '2026-02-11', '10:30:00', 1, NULL, 5004, '', NULL, 2, '2026-02-10', '10004',
2026, 4), (2, '2026-02-10', 1005, '1995-07-14', 'CP-005', 'Carlos Miguel Tan', 'Dr. Elena
Villanueva', '654 Luna St, Makati', 'HCP-005', 'Tan', 'Carlos', 'Miguel', 'SPA-005', 'P005',
'BK003', 'SR005', 'Atty. Dela Rosa', '2025-12-15', 0, NULL, NULL, 3, NULL, 5005, '', NULL, 1,
'2026-02-10', '10005', 2026, 5), (2, '2026-02-11', 1006, '1982-01-08', 'CP-006', 'Sofia Isabelle
Lim', 'Dr. Ramon Bautista', '987 Silang Ave, Pasig', 'HCP-006', 'Lim', 'Sofia', 'Isabelle',
'SPA-006', 'P006', 'BK003', 'SR006', 'Atty. Mercado', '2025-12-20', 0, NULL, NULL, 1, NULL, 5006,
'', NULL, 2, '2026-02-11', '10006', 2026, 6), (3, '2026-03-05', 1007, '1998-09-18', 'CP-007',
'Rafael Antonio Uy', 'Dr. Patricia Ong', '159 Roxas Blvd, Paranaque', 'HCP-007', 'Uy', 'Rafael',
'Antonio', 'SPA-007', 'P007', 'BK004', 'SR007', 'Atty. Garcia', '2026-01-05', 1, '2026-03-06',
'14:15:00', 2, NULL, 5007, '', NULL, 1, '2026-03-05', '10007', 2026, 7), (3, '2026-03-05', 1008,
'1991-12-03', 'CP-008', 'Gabriela Nicole Sy', 'Dr. Fernando Aquino', '753 Quezon Blvd, Taguig',
'HCP-008', 'Sy', 'Gabriela', 'Nicole', 'SPA-008', 'P008', 'BK004', 'SR008', 'Atty. Ramos',
'2026-01-10', 0, NULL, NULL, 1, NULL, 5008, '', NULL, 2, '2026-03-05', '10008', 2026, 8), (3,
'2026-03-06', 1009, '1987-04-22', 'CP-009', 'Miguel Alfonso Co', 'Dr. Teresa Javier', '246 Osmena
St, Baguio', 'HCP-009', 'Co', 'Miguel', 'Alfonso', 'SPA-009', 'P009', 'BK005', 'SR009', 'Atty.
Villar', '2026-01-15', 0, NULL, NULL, 3, NULL, 5009, 'III', NULL, 1, '2026-03-06', '10009', 2026,
9), (4, '2026-04-20', 1010, '1993-06-11', 'CP-010', 'Isabella Marie Chua', 'Dr. Roberto Domingo',
'135 Kalayaan Ave, Pampanga', 'HCP-010', 'Chua', 'Isabella', 'Marie', 'SPA-010', 'P010', 'BK005',
'SR010', 'Atty. Pineda', '2026-02-01', 0, NULL, NULL, 1, NULL, 5010, '', NULL, 2, '2026-04-20',
'10010', 2026, 10), (4, '2026-04-20', 1011, '1980-10-05', 'CP-011', 'Antonio Luis Pangilinan',
'Dr. Carmen Salazar', '864 EDSA, Mandaluyong', 'HCP-011', 'Pangilinan', 'Antonio', 'Luis',
'SPA-011', 'P011', 'BK006', 'SR011', 'Atty. Hernandez', '2026-02-10', 1, '2026-04-21', '09:00:00',
2, NULL, 5011, '', NULL, 1, '2026-04-20', '10011', 2026, 11);
```

Version: 1.3
Date: September 7, 2026
Purpose: Sync bridge between Progress OpenEdge and MySQL for Entrust Instant ID card printing. Reads `"id-print"` from OpenEdge and mirrors it to MySQL's `id_print` table.

Architecture Overview

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ Progress OpenEdge SQL Server ■ ← Source of truth
(remote, provided by client) ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ ODBC
■■■■■■■■■■■■■■■■■▼■■■■■■■■■■■■■■■■■■■ ■ OpenEdge ODBC Driver ■ ←
PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe ■ (DataDirect 8.0, 64-bit) ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ ■■■■■■■■■■■■■■■■■▼■■■■■■■■■■■■■■■■■■■ ■ C# Windows
Service ■ ← OpenEdgeBridge.exe (this application) ■ (.NET 8, self-contained) ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ MySQL Connector
```

```
■■■■■■■■■■■■■■■■▼■■■■■■■■■■■■■■■■■■ ■ MySQL Server (XAMPP) ■ ← Local database, runs on the same PC ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ JDBC (mysql-connector-j-8.0.33.jar)
■■■■■■■■■■■■■■■■▼■■■■■■■■■■■■■■■■■■ ■ Entrust Instant ID ■ ← Card issuance system
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

Data Flow

OpenEdge → MySQL (One-Way Sync)

```
OpenEdge "id-print" ■■■ C# Service ■■■ MySQL id_print table (read) (sync loop) (upsert)
```

All 31 fields from the `id-print` table are synced, including:
- `batch-no`, `batch-date`, `app-no`, `dob`
- `cp-no`, `fullname`, `hcp-fullname`, `address`, `hcp-cpno`
- `lname`, `fname`, `mname`
- `spa-docno`, `page-no`, `book-no`, `series-no`, `spa-atty`, `spa-dt-issue`
- `printed`, `printed-dt`, `printed-tm`
- `tran-type`, `image-data`, `id-code`, `sfx`, `photo`
- `gender`, `date-issued`, `id-no`, `year-applied`, `control-no`

Runs every N seconds (configurable in `appsettings.json`). Uses **check-then-update** upserts (SELECT by `app_no`, then UPDATE or INSERT) to avoid duplicates. `print_count` is only inserted on first sync; subsequent updates do NOT overwrite it (Instant ID manages `print_count`).

Full vs. Incremental Sync

The bridge has two read modes, decided automatically on every cycle:

Mode	Trigger	What it reads from OpenEdge
FULL	No `last_full_sync_utc` recorded yet (first run, or fresh MySQL)	All rows in `pub."id-print"` — seeds/refreshes the entire MySQL table
INCREMENTAL	A full sync has already been recorded	Only rows with `batch-no` greater than the highest `batch_no` already in MySQL

OpenEdge `id-print` records are **write-once** — new data arrives only as new batches; existing rows are never edited. Therefore, with the default `Sync:FullSyncIntervalMinutes = 0`, the full sync runs **once** (the initial seed) and every later cycle is incremental. Restarting the service or rebooting the PC does **not** cause another full sync — the timestamp is persisted in MySQL's `sync_state` table (see Database Tables).

If `Sync:FullSyncIntervalMinutes` is set to a positive value (e.g., `60`), a full sync also runs when that many minutes have elapsed since the last successful full sync — useful only if OpenEdge rows ever start getting edited in place.

> **Note:** `signature_data` and `qr_data` are **not** part of this sync — they are MySQL-only columns populated by Entrust Instant ID and never touched by the bridge (see Database Tables).

MySQL → OpenEdge (Printed Back-Sync)

```
MySQL id_print ■■■■■■■■ C# Service ■■■■■■■■ OpenEdge "id-print" (print_count > 0) (check + mark) (SET printed = yes)
```

When Instant ID increments `print_count > 0` on a record, the bridge detects it and sets `printed = yes` in the corresponding OpenEdge record.

Prerequisites

On the Target Windows PC

#	Component	Version / File	Notes
1	**Windows OS**	Windows 10/11 or Windows Server 2016+	64-bit required
2	**OpenEdge ODBC Driver**	`PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe`	Install, configure a System DSN
3	**XAMPP**	Latest (8.x)	Provides local MySQL server
4	**MySQL Connector/J**	`mysql-connector-j-8.0.33.jar`	For Entrust Instant ID to connect to MySQL
5	**OpenEdgeBridge**	Published `publish/` folder	Self-contained, no .NET runtime needed

---

Local Development — Installing Progress OpenEdge Database

> **Use this section when no remote OpenEdge server is available.**
> You can install Progress OpenEdge locally, create a database, and run the SQL below.

Step 0-A: Download & Install Progress OpenEdge

1. Go to https://www.progress.com/openedge/downloads
2. Download **Progress OpenEdge Developer Studio** or **Progress OpenEdge Personal Edition** (free for development)
3. Run the installer (typically `PROGRESS_OE_12.x_WIN_64.exe`) as Administrator
4. Choose **Typical** or **Complete** installation
5. Default install path: `C:\Progress\OpenEdge\`

After installation, a **"Proenv"** shortcut will appear in your Start Menu.
Proenv is a command prompt pre-configured with all OpenEdge environment variables.

Step 0-B: Create a Local OpenEdge Database

Open **Proenv** as Administrator and run:

```
:: Create a folder for your database mkdir C:\OpenEdgeDatabases cd C:\OpenEdgeDatabases :: Create an empty database called 'iddb' prodb iddb empty
```

This creates `iddb.db` and related files in `C:\OpenEdgeDatabases\`.

Step 0-C: Start the Database Server

In **Proenv**, run:

```
:: Start the database for multi-user access with both 4GL and SQL enabled proserve "C:\OpenEdgeDatabases\iddb.db" -S 11918 -ServerType Both -n 51 -L 8192
```

Parameter	Meaning
`-S 11918`	Service port (SQL and 4GL clients connect on this port)
`-n 51`	Maximum concurrent users
`-L 8192`	Lock table size (entries)
`-ServerType Both`	**Required** — enables both 4GL and SQL-92 engine

The database is now running. Keep this Proenv window open.

Step 0-D: Configure the ODBC System DSN

1. Press `Win + R`, type `odbcad32.exe`, press Enter
2. Go to the **System DSN** tab, click **Add...**
3. Select **Progress OpenEdge 12.x Driver** (or "DataDirect OpenEdge SQL Server")
4. Configure:
5. **Data Source Name:** `OpenEdgeInstantID`
6. **Description:** `OpenEdge Instant ID Database`
7. **Host Name:** `localhost`
8. **Port Number:** `11918`
9. **Database Name:** `iddb`
10. **User ID:** `admin`
11. Click **Test Connect** → should succeed
12. Click **OK**

Step 0-E: Create the Database Schema (`.df` File)

The table schema is defined in a `.df` (Data Definition) file. Create a file named `iddb.df` in `C:\OpenEdgeDatabases\` with the following content:

```
ADD TABLE "id-print" AREA "Schema Area" DUMP-NAME "id-print" ADD FIELD "batch-no" OF "id-print" AS
integer FORMAT "->,>>>,>>9" INITIAL "0" POSITION 2 MAX-WIDTH 4 ORDER 10 ADD FIELD "batch-date" OF
"id-print" AS date FORMAT "99/99/9999" INITIAL ? POSITION 3 MAX-WIDTH 4 ORDER 20 ADD FIELD
"date-issued" OF "id-print" AS date FORMAT "99/99/9999" INITIAL ? POSITION 29 MAX-WIDTH 4 ORDER
280 ADD FIELD "app-no" OF "id-print" AS integer FORMAT "->,>>>,>>9" INITIAL "0" POSITION 4
MAX-WIDTH 4 ORDER 30 ADD FIELD "dob" OF "id-print" AS date FORMAT "99/99/9999" INITIAL ? POSITION
5 MAX-WIDTH 4 ORDER 40 ADD FIELD "cp-no" OF "id-print" AS character FORMAT "x(15)" INITIAL ""
POSITION 6 MAX-WIDTH 30 ORDER 50 ADD FIELD "fullname" OF "id-print" AS character FORMAT "x(65)"
INITIAL "" POSITION 7 MAX-WIDTH 130 ORDER 60 ADD FIELD "hcp-fullname" OF "id-print" AS character
FORMAT "x(65)" INITIAL "" POSITION 8 MAX-WIDTH 130 ORDER 70 ADD FIELD "address" OF "id-print" AS
character FORMAT "x(65)" INITIAL "" POSITION 9 MAX-WIDTH 130 ORDER 80 ADD FIELD "hcp-cpno" OF
"id-print" AS character FORMAT "x(15)" INITIAL "" POSITION 10 MAX-WIDTH 30 ORDER 90 ADD FIELD
"lname" OF "id-print" AS character FORMAT "x(55)" INITIAL "" POSITION 11 MAX-WIDTH 110 ORDER 100
ADD FIELD "fname" OF "id-print" AS character FORMAT "x(55)" INITIAL "" POSITION 12 MAX-WIDTH 110
ORDER 110 ADD FIELD "mname" OF "id-print" AS character FORMAT "x(55)" INITIAL "" POSITION 13
MAX-WIDTH 110 ORDER 120 ADD FIELD "spa-docno" OF "id-print" AS character FORMAT "x(25)" INITIAL ""
POSITION 14 MAX-WIDTH 50 ORDER 130 ADD FIELD "page-no" OF "id-print" AS character FORMAT "x(20)"
INITIAL "" POSITION 15 MAX-WIDTH 40 ORDER 140 ADD FIELD "book-no" OF "id-print" AS character
FORMAT "x(20)" INITIAL "" POSITION 16 MAX-WIDTH 40 ORDER 150 ADD FIELD "series-no" OF "id-print"
AS character FORMAT "x(20)" INITIAL "" POSITION 17 MAX-WIDTH 40 ORDER 160 ADD FIELD "spa-atty" OF
"id-print" AS character FORMAT "x(65)" INITIAL "" POSITION 18 MAX-WIDTH 130 ORDER 170 ADD FIELD
"spa-dt-issue" OF "id-print" AS date FORMAT "99/99/9999" INITIAL ? POSITION 19 MAX-WIDTH 4 ORDER
180 ADD FIELD "printed" OF "id-print" AS logical FORMAT "yes/no" INITIAL "no" POSITION 20
MAX-WIDTH 1 ORDER 190 ADD FIELD "printed-dt" OF "id-print" AS date FORMAT "99/99/9999" INITIAL ?
POSITION 21 MAX-WIDTH 4 ORDER 200 ADD FIELD "printed-tm" OF "id-print" AS character FORMAT "x(35)"
INITIAL "" POSITION 22 MAX-WIDTH 70 ORDER 210 ADD FIELD "tran-type" OF "id-print" AS integer
DESCRIPTION "1 = for RTC, 2=senior" FORMAT "->,>>>,>>9" INITIAL "0" POSITION 23 MAX-WIDTH 4 ORDER
220 ADD FIELD "image-data" OF "id-print" AS raw FORMAT "x(55)" INITIAL "" POSITION 24 MAX-WIDTH
110 ORDER 230 ADD FIELD "id-code" OF "id-print" AS integer FORMAT "->,>>>,>>9" INITIAL "0"
POSITION 25 MAX-WIDTH 4 ORDER 240 ADD FIELD "sfx" OF "id-print" AS character FORMAT "x(15)"
INITIAL "" POSITION 26 MAX-WIDTH 30 ORDER 250 ADD FIELD "photo" OF "id-print" AS blob FORMAT
"x(8)" INITIAL ? POSITION 27 LOB-AREA "Schema Area" LOB-BYTES 104857600 LOB-SIZE 100M ORDER 260
ADD FIELD "gender" OF "id-print" AS integer DESCRIPTION "1=Female, 2=Male" FORMAT "->,>>>,>>9"
INITIAL "0" POSITION 28 MAX-WIDTH 4 ORDER 270 ADD FIELD "year-applied" OF "id-print" AS integer
FORMAT ">>>9999" INITIAL "0" POSITION 31 MAX-WIDTH 4 ORDER 300 ADD FIELD "control-no" OF
"id-print" AS integer FORMAT "->,>>>,>>9" INITIAL "0" POSITION 32 MAX-WIDTH 4 ORDER 310 ADD FIELD
"id-no" OF "id-print" AS character FORMAT "x(25)" INITIAL "" POSITION 33 MAX-WIDTH 50 ORDER 320
ADD INDEX "rtcid-idx" ON "id-print" AREA "Schema Area" PRIMARY INDEX-FIELD "batch-no" ASCENDING
INDEX-FIELD "app-no" ASCENDING INDEX-FIELD "tran-type" ASCENDING ADD INDEX "oscaid-idx" ON
"id-print" AREA "Schema Area" INDEX-FIELD "control-no" ASCENDING INDEX-FIELD "id-no" ASCENDING
INDEX-FIELD "tran-type" ASCENDING . PSC cpstream=ISO8859-1 . 0000004144
```

Then, in **Proenv**, load the `.df` file into the database:

```
cd C:\OpenEdgeDatabases prodict iddb
```

In the Data Dictionary menu, select **Admin** → **Load Data Definitions** → **.df file**, then choose `iddb.df`.

> **Note:** The database name is `iddb` and the schema resides under the `PUB` area (default).

Database Tables

`id-print` / `id_print`

> Converted from the provided `.df` schema.
> **Paste each CREATE TABLE directly into your SQL editor and run it.**

OpenEdge SQL — CREATE TABLE

```
CREATE TABLE "id-print" ( "batch-no" INTEGER NOT NULL DEFAULT 0, "batch-date" DATE NULL, "app-no"
INTEGER NOT NULL DEFAULT 0, "dob" DATE NULL, "cp-no" VARCHAR(15) NOT NULL DEFAULT '', "fullname"
VARCHAR(65) NOT NULL DEFAULT '', "hcp-fullname" VARCHAR(65) NOT NULL DEFAULT '', "address"
VARCHAR(65) NOT NULL DEFAULT '', "hcp-cpno" VARCHAR(15) NOT NULL DEFAULT '', "lname" VARCHAR(55)
NOT NULL DEFAULT '', "fname" VARCHAR(55) NOT NULL DEFAULT '', "mname" VARCHAR(55) NOT NULL DEFAULT
'', "spa-docno" VARCHAR(25) NOT NULL DEFAULT '', "page-no" VARCHAR(20) NOT NULL DEFAULT '',
"book-no" VARCHAR(20) NOT NULL DEFAULT '', "series-no" VARCHAR(20) NOT NULL DEFAULT '', "spa-atty"
VARCHAR(65) NOT NULL DEFAULT '', "spa-dt-issue" DATE NULL, "printed" BIT NOT NULL DEFAULT 0,
"printed-dt" DATE NULL, "printed-tm" VARCHAR(35) NOT NULL DEFAULT '', "tran-type" INTEGER NOT NULL
DEFAULT 0, "image-data" VARBINARY(110) NULL, "id-code" INTEGER NOT NULL DEFAULT 0, "sfx"
VARCHAR(15) NOT NULL DEFAULT '', "photo" BLOB NULL, "gender" INTEGER NOT NULL DEFAULT 0,
"date-issued" DATE NULL, "id-no" VARCHAR(25) NOT NULL DEFAULT '', "year-applied" INTEGER NOT NULL
DEFAULT 0, "control-no" INTEGER NOT NULL DEFAULT 0 );
```

> **Note:** OpenEdge requires double-quotes around field names containing hyphens (`-`).
> The schema resides in the `PUB` area (default).
> `BIT` represents the OpenEdge `LOGICAL` type (yes/no → 1/0).
> `VARBINARY(110)` stores the raw image binary data (MAX-WIDTH 110 from the schema).
> `BLOB` stores the cardholder photo (LOB-SIZE 100M from the schema).

MySQL — CREATE TABLE

```
CREATE TABLE IF NOT EXISTS id_print ( id INT NOT NULL AUTO_INCREMENT, batch_no INT NULL,
batch_date DATE NULL, app_no INT NULL, dob DATE NULL, cp_no VARCHAR(15) NULL, fullname VARCHAR(65)
NULL, hcp_fullname VARCHAR(65) NULL, address VARCHAR(100) NULL, hcp_cpno VARCHAR(15) NULL, lname
VARCHAR(55) NULL, fname VARCHAR(55) NULL, mname VARCHAR(55) NULL, spa_docno VARCHAR(25) NULL,
page_no VARCHAR(20) NULL, book_no VARCHAR(20) NULL, series_no VARCHAR(20) NULL, spa_atty
VARCHAR(65) NULL, spa_dt_issue DATE NULL, printed VARCHAR(3) NULL COMMENT 'yes / no (matches
OpenEdge LOGICAL FORMAT)', printed_dt DATE NULL, printed_tm VARCHAR(35) NULL, tran_type INT NULL
COMMENT '1 = for RTC, 2 = senior', image_data LONGBLOB NULL, id_code INT NULL, sfx VARCHAR(15)
NULL, photo LONGBLOB NULL, gender INT NULL COMMENT '1 = Female, 2 = Male', date_issued DATE NULL,
id_no VARCHAR(25) NULL, year_applied INT NULL, control_no INT NULL, signature_data LONGBLOB NULL
COMMENT 'Signature image; written by Entrust Instant ID (MySQL only, NOT synced from OpenEdge)',
qr_data LONGBLOB NULL COMMENT 'QR code image; written by Entrust Instant ID (MySQL only, NOT
synced from OpenEdge)', print_count INT NULL COMMENT 'Incremented by Instant ID; triggers
printed=yes back-sync', PRIMARY KEY (id), UNIQUE INDEX idx_app_no (app_no), INDEX idx_batch_no
(batch_no), INDEX idx_print_sync (printed, print_count, app_no) ) ENGINE=InnoDB DEFAULT
CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

> **Note:** `id` is an internal auto-increment primary key. `app_no` is the unique business key — the bridge uses it for deduplication (it SELECTs by `app_no` first, then UPDATEs or INSERTs as needed).

> **MySQL-only columns (`signature_data`, `qr_data`):** These store the cardholder's signature and QR-code images and are **MySQL only**. They do **not** exist in the OpenEdge `"id-print"` table and are **never read from or written back to OpenEdge**. Entrust Instant ID populates them directly in MySQL. The bridge deliberately leaves them untouched — it does

not include them in its INSERT/UPDATE, so syncing from OpenEdge will never overwrite them.

Already have the table? If you created `id_print` before these columns existed, add them with:

```
sql ALTER TABLE id_print MODIFY COLUMN address VARCHAR(100) NULL, CHANGE COLUMN id_type tran_type INT NULL COMMENT '1 = for RTC, 2 = senior', ADD COLUMN image_data LONGBLOB NULL AFTER tran_type, ADD COLUMN sfx VARCHAR(15) NULL AFTER id_code, ADD COLUMN photo LONGBLOB NULL AFTER sfx, ADD COLUMN gender INT NULL COMMENT '1 = Female, 2 = Male' AFTER photo, ADD COLUMN date_issued DATE NULL AFTER gender, MODIFY COLUMN id_no VARCHAR(25) NULL AFTER gender, ADD COLUMN year_applied INT NULL AFTER id_no, ADD COLUMN control_no INT NULL AFTER year_applied, ADD COLUMN signature_data LONGBLOB NULL AFTER control_no, ADD COLUMN qr_data LONGBLOB NULL AFTER signature_data, ADD INDEX idx_batch_no (batch_no), ADD INDEX idx_print_sync (printed, print_count, app_no);
```

Schema update (2026-09-07): OpenEdge now stores `id-no` as character and adds the `rtcid-idx` and `oscaid-idx` indexes. `date-issued` remains part of both OpenEdge and MySQL; `id_no` is stored as `VARCHAR(25)` in MySQL. Reload the updated `.df` (Step 0-E), then run the ALTER above on MySQL.

`sync_state` (bridge state table — MySQL only, auto-created)

The bridge keeps its own bookkeeping in a tiny table called `sync_state`. **You do not need to create it manually** — the bridge runs `CREATE TABLE IF NOT EXISTS` on demand, so it appears in MySQL automatically after the first sync cycle.

```
CREATE TABLE IF NOT EXISTS sync_state ( state_key VARCHAR(64) NOT NULL PRIMARY KEY, state_value VARCHAR(64) NOT NULL, updated_utc DATETIME NOT NULL ) ENGINE=InnoDB;
```

Key	Meaning
`last_full_sync_utc`	UTC timestamp of the last successful FULL sync. Used to decide whether the next cycle can be INCREMENTAL. Survives restarts, so a reboot never forces a redundant full sync.

Column Comparison — Plain English

Think of each row as **one person's ID-card application folder**. The columns are the details written on that folder.

OpenEdge name	MySQL name	Plain English meaning
`batch-no`	`batch_no`	**Group number** — which batch/run of applications this person came in with (like a batch printing number)
`batch-date`	`batch_date`	**Date of that group** — when their batch was made
`app-no`	`app_no`	**Application number** — the unique file number of this person's ID application. This is the "fingerprint" that tells us which person is which
`dob`	`dob`	**Date of birth**

`cp-no`	`cp_no`	**The person's reference number** (their own control/card number)
`fullname`	`fullname`	**Full name** — complete name all in one
`hcp-fullname`	`hcp_fullname`	**Doctor's name** — the healthcare provider attached to this person
`address`	`address`	**Home address**
`hcp-cpno`	`hcp_cpno`	**Doctor's number** — the healthcare provider's own reference number
`lname`	`lname`	**Last name** (surname)
`fname`	`fname`	**First name**
`mname`	`mname`	**Middle name**
`spa-docno`	`spa_docno`	**Permission letter number** — the number of the Special Power of Attorney (SPA) document
`spa-atty`	`spa_atty`	**Lawyer's name** — the attorney who signed that permission letter
`spa-dt-issue`	`spa_dt_issue`	**Date the permission letter was issued**
`page-no`	`page_no`	**Page number** — which page in the record book this person is on
`book-no`	`book_no`	**Book number** — which record book
`series-no`	`series_no`	**Series number** — the serial number of the ID
`tran-type`	`tran_type`	**Kind of ID / transaction type** — `1` = regular (RTC), `2` = senior citizen
`image-data`	`image_data`	**The person's photo** (picture file)
`id-code`	`id_code`	**ID code** — the code number printed on the card
`sfx`	`sfx`	**Name suffix** — e.g. Jr., Sr., III
`photo`	`photo`	**Photo** — the cardholder's photograph (BLOB image)
`gender`	`gender`	**Gender** — `1` = Female, `2` = Male
`date-issued`	`date_issued`	**Date the ID card was issued**
`id-no`	`id_no`	**ID number** — character value from OpenEdge

`year-applied`	`year_applied`	**Year the application was filed** (e.g. `2026`)
`control-no`	`control_no`	**Control number** — application/control tracking number
`printed`	`printed`	**Has it been printed yet?** — `yes` or `no`
`printed-dt`	`printed_dt`	**Date it was printed**
`printed-tm`	`printed_tm`	**Time it was printed**
—	`signature_data`	**Person's signature picture** — *MySQL only*. Not in OpenEdge, not synced back. Entrust fills this in
—	`qr_data`	**QR code picture** — *MySQL only*. Not in OpenEdge, not synced back. Entrust fills this in
—	`print_count`	**How many times it got printed** — *MySQL only*. Instant ID counts this; it's never overwritten by the bridge
—	`id`	**Row number** — MySQL's own internal counter. Not business data, ignore it

30-second summary: The bridge copies the person's details (name, birthdate, address, photo, etc.) **from OpenEdge → MySQL**, like photocopying the folder. `app_no` is the **matchmaker** — it makes sure updates land on the right person. The three columns that exist **only in MySQL** (`signature_data`, `qr_data`, `print_count`) are filled in by the **Entrust machine**, and the bridge never touches them.

Step-by-Step Setup (Remote Server)

> **If you already have a remote OpenEdge server**, follow these steps.
> **If no server is available**, go back to Local Development — Installing Progress OpenEdge Database first.

1. Install OpenEdge ODBC Driver

Run the installer as Administrator:

```
PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe
```

After installation, open **ODBC Data Sources (64-bit)** and create a **System DSN**:

1. Press `Win + R`, type `odbcad32.exe`, press Enter
2. Go to the **System DSN** tab
3. Click **Add...**

4. Select **Progress DataDirect OpenEdge SQL Server** (or similar)
5. Configure:
6. **Data Source Name:** e.g., `OpenEdgeInstantID`
7. **Host:** (provided by your OpenEdge team, or `localhost` if local)
8. **Port:** (provided by your OpenEdge team, or `11918` if local)
9. **Database:** (provided by your OpenEdge team, or `iddb` if local)
10. **User ID / Password:** (provided by your OpenEdge team)
11. Click **Test Connect** to verify, then **OK**

2. Create the OpenEdge Tables

Connect to your OpenEdge database using any ODBC-compatible SQL client (DBeaver, HeidiSQL, etc.) and paste the SQL from the Database Tables section above:

- Run the `"id-print"` CREATE TABLE

2-B. Grant Table Permissions to the Login User

The bridge connects to OpenEdge using the **User ID** configured in the ODBC DSN (e.g., `admin`). If your login user is **not** `admin`, it needs explicit permissions on the table — otherwise the bridge fails with an access-denied error when it starts.

Run the following from a DBA account (e.g., `admin`), replacing `your_login_user` with the actual user you put in the DSN:

```
GRANT SELECT, INSERT, UPDATE, DELETE ON pub."id-print" TO your_login_user;
```

> **Note:** `SELECT` is required to read records for the OpenEdge → MySQL sync, and `UPDATE` is required for the printed back-sync (`printed = yes`). `INSERT`/`DELETE` are included for completeness. `pub` is the default schema area, and the double quotes around `"id-print"` are required because of the hyphen in the table name.

3. Install XAMPP

1. Download from https://www.apachefriends.org/
2. Run the installer with default settings
3. Open the **XAMPP Control Panel**
4. Click **Start** next to **MySQL**
5. Click **Admin** to open phpMyAdmin

4. Create the MySQL Database & Tables

In phpMyAdmin (or any MySQL client), run:

```
CREATE DATABASE IF NOT EXISTS instantid_bridge CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE instantid_bridge;
```

Then paste and run the MySQL CREATE TABLE statements from the Database Tables section above:

- Run the **`id_print`** CREATE TABLE

5. Configure Entrust Instant ID (MySQL Connector/J)

1. Copy `mysql-connector-j-8.0.33.jar` to Entrust's `lib/` folder
2. Configure Entrust to point to the local MySQL database:
3. **Host:** `localhost`
4. **Port:** `3306`
5. **Database:** `instantid_bridge`
6. **User:** `root` (or a dedicated user)
7. **Table:** `id_print`

6. Deploy the Bridge Service

Copy the entire `publish/` folder to the target PC, e.g. `C:\OpenEdgeBridge\`.

Edit `appsettings.json`

```
{ "ConnectionStrings": { "OpenEdge": "DSN=OpenEdgeInstantID;Uid=your_user;Pwd=your_password;", "MySql": "Server=localhost;Database=instantid_bridge;Uid=root;Pwd=your_mysql_password;" }, "Sync": { "IntervalSeconds": 10, "FullSyncIntervalMinutes": 0 } }
```

Setting	Description
`OpenEdge`	ODBC DSN name, user, and password
`MySql`	MySQL server, database, user, and password
`Sync:IntervalSeconds`	How often the sync loop runs (default: 10s)
`Sync:FullSyncIntervalMinutes`	Minutes between periodic FULL syncs. `0` = disabled (recommended; OpenEdge rows are write-once, so only the first run seeds). Positive value (e.g., `60`) = re-read the whole table that often, catching in-place edits if they ever happen

Test the executable manually

Before registering the service, open PowerShell and run the published executable from its deployment folder:

```
Set-Location C:\OpenEdgeBridge .\OpenEdgeBridge.exe
```

The bridge runs in the foreground and writes startup, connection, and sync errors to the PowerShell window. Confirm that rows appear in `instantid_bridge.id_print`, then press **Ctrl+C** to stop it.

> Stop the manual test before starting the Windows service so both processes do not run at the same time.

7. Install as a Windows Service

Open **PowerShell as Administrator** and run:

```
New-Service -Name "OpenEdgeBridge" ` -BinaryPathName "C:\OpenEdgeBridge\OpenEdgeBridge.exe" ` -DisplayName "OpenEdge Bridge Service" ` -Description "Syncs id-print data between OpenEdge and MySQL for Entrust Instant ID" ` -StartupType Automatic Start-Service OpenEdgeBridge
```

Verify:

```
Get-Service OpenEdgeBridge # Status should be: Running
```

Service Management Commands

```
Start-Service OpenEdgeBridge # Start Stop-Service OpenEdgeBridge # Stop Restart-Service OpenEdgeBridge # Restart (after config changes) Get-Service OpenEdgeBridge # Check status # Uninstall Stop-Service OpenEdgeBridge sc delete OpenEdgeBridge
```

Logging & Troubleshooting

Log Locations

Source	Location
File logs	`C:\OpenEdgeBridge\Logs\` (dated log files)
Windows Event Viewer	*Windows Logs* → *Application*, Source: `OpenEdgeBridge`

Common Issues

Problem	Likely Cause	Fix
Service starts then stops immediately	Wrong DSN name or credentials	Test the ODBC DSN in ODBC Administrator with **Test Connect**
"MySQL connection string not found"	`appsettings.json` not in same folder as `.exe`	Ensure `appsettings.json` is in `C:\OpenEdgeBridge\`
Duplicate rows in MySQL	Missing UNIQUE index	Run the `ALTER TABLE ... ADD UNIQUE INDEX` SQL above
OpenEdge ODBC driver not found	Wrong bitness (32 vs 64)	Use `odbcad32.exe` (64-bit), not the 32-bit one in SysWOW64

`Failure getting schema read lock (12138)`	SQL engine not initialized for database	1. Verify `-ServerType Both` or `-ServerType SQL` on proserve 2. In sqlexp run: `UPDATE TABLE STATISTICS AND INDEX STATISTICS AND ALL COLUMN STATISTICS;` 3. If still failing, restart DB: `proshut` then `proserve` with `-SQLTrc 3`
MySQL not accessible from Entrust	MySQL not running or wrong port	Check XAMPP Control Panel; MySQL must be **Started**
OpenEdge SQL CREATE TABLE fails with "Invalid table name"	Hyphens in table/field names not quoted	Wrap names containing hyphens in double quotes: `"id-print"`, `"batch-no"`
`image-data` field too small	VARBINARY length insufficient	The OpenEdge schema specifies MAX-WIDTH 110; adjust as needed
Full sync on every start (log says `[FULL]`)	`sync_state` missing or MySQL unreachable at startup	The bridge auto-creates `sync_state`; after the first successful full sync it records a timestamp and later starts use `[INCREMENTAL]`. If MySQL is down at startup it falls back to FULL — that is safe and self-corrects once MySQL is reachable

---

Sync Interval Tuning

`IntervalSeconds` — how often the bridge checks for new batches

The default sync interval is **10 seconds**. Adjust in `appsettings.json`:

```
"Sync": { "IntervalSeconds": 30 }
```

- **Lower** (e.g., 5s) = faster sync, more database load
- **Higher** (e.g., 60s) = slower sync, less load
- Recommended: 10–30 seconds for production

The interval is an **idle pause between cycles** (work → wait → work), not a fixed wall-clock schedule. A slow sync simply delays the next check; cycles never overlap.

`FullSyncIntervalMinutes` — how often the whole table is re-read

```
"Sync": { "FullSyncIntervalMinutes": 0 }
```

Value	Behavior

`0` (default, recommended)	FULL sync only on the **first run** (seeds MySQL). All later cycles are INCREMENTAL (`batch-no` > highest in MySQL). Restarts never re-trigger a full sync.
Positive (e.g., `60`)	Additionally re-reads the entire table when that many minutes have passed since the last full sync. Only needed if OpenEdge rows can be edited in place.

How the decision works: after every successful FULL sync the bridge writes the completion time to `sync_state.last_full_sync_utc` in MySQL. On each cycle it compares that timestamp against `FullSyncIntervalMinutes`. If MySQL is unreachable at startup, the bridge falls back to FULL — the safe default that guarantees a complete copy.

Why `0` is right for this deployment: OpenEdge `id-print` data is batch-based and write-once — new records always arrive in new batches (higher `batch-no`), and existing rows are not edited. The incremental query therefore sees every new record, and a periodic full re-read only wastes database load on a table that never changes. If the provider ever starts correcting existing rows, set this to `60` (hourly refresh) or similar.

Port Summary

Component	Port	Direction
OpenEdge SQL Server	Provided by OpenEdge team	Outbound
MySQL (XAMPP)	3306	Local
Entrust → MySQL	3306	Local via JDBC

No inbound ports are opened by the bridge itself.

Upgrade Procedure

1. Stop the service: `Stop-Service OpenEdgeBridge`
2. Replace all files in `C:\OpenEdgeBridge\` with the new `publish/` output
3. Preserve your `appsettings.json` (do not overwrite with default)
4. Start the service: `Start-Service OpenEdgeBridge`

> **After upgrading from v1.1 or earlier:** the new `sync_state` table does not exist yet, so the first run performs one FULL sync to seed MySQL and record the timestamp. From then on, only new batches are synced — including across restarts.

Checklist for Target PC

- [ ] Windows 10/11 or Server 2016+ (64-bit)

- [ ] OpenEdge ODBC driver installed and DSN configured
- [ ] XAMPP installed and MySQL running
- [ ] `instantid_bridge` database created in MySQL
- [ ] `id-print` / `id_print` table created in **both** OpenEdge and MySQL
- [ ] `sync_state` table present in MySQL (auto-created by the bridge after the first run)
- [ ] `mysql-connector-j-8.0.33.jar` placed in Entrust's lib folder
- [ ] Bridge `publish/` folder copied to `C:\OpenEdgeBridge\`
- [ ] `appsettings.json` updated with correct connection strings
- [ ] Windows Service installed and started
- [ ] Logs checked for errors